

Zero-shot RS-VLMs vs. our trained model

Family-1 overhead urban-attribute benchmark — satellite-only, $n = 3,135$

What this compares

Four off-the-shelf **remote-sensing VLMs**, evaluated **zero-shot** on the **shared benchmark, satellite-only**, against **our fine-tuned Qwen3.5-9B**.

- All five see the **same single marked satellite tile** (our sat mode), the **same 9 Family-1 tasks**, the **same 3,135 questions** — like-for-like input.
- The four externals are **single-image, overhead-only** and **never trained** on EOLLM, on our 4-way MCQ format, or on the red-dot marker our tiles carry $\Rightarrow$ pure out-of-distribution transfer.
- Fair comparison column = **our trained sat** (single satellite view, identical input modality).

All: greedy, max_new_tokens=8, single GPU (RTX PRO 6000 Blackwell), batched.

The four external models

Model	LLM / vision	Image preprocessing
GeoChat-7B	Vicuna-7B / CLIP-ViT-L	504px single crop
SkySenseGPT-7B	Vicuna-7B / CLIP-ViT-L	504px single crop
LHRS-Bot-Nova	Llama-3-8B / SigLIP-384 + perceiver	384px single
EarthDial-4B-RGB	Phi-3-Mini / InternViT-300M	InternVL $\leq 6 \times 448\text{px}$ + thumb
Ours: Qwen3.5-9B (LoRA)	Qwen3.5-9B / native ViT	trained on EOLLM (sat mode)

LHRS-Bot & EarthDial loaded via their native repos (custom perceiver / `trust_remote_code` InternVL); GeoChat/SkySenseGPT via the original-LLaVA loader. No flash-attention (Blackwell sm_120): SDPA / eager fallback.

Overall Family-1 accuracy (sat-only, $n = 3,135$)

Model	Overall acc
GeoChat-7B	27.8
LHRS-Bot-Nova	32.9
SkySenseGPT-7B	33.4
EarthDial-4B-RGB	41.3 (best external)
Ours — trained sat (per-city)	54.9
Ours — trained sat (seen/unseen)	54.9

Best zero-shot RS model (EarthDial) trails our trained sat by **13.6 pts**; GeoChat by **27.1 pts**.

Same single satellite view for all.

Per-task: all four externals vs. our trained sat

task	GeoChat	SkySense	LHRS	EarthDial	pc sat	su sat
Land use	48.7	53.4	46.3	47.5	75.5	72.0
Building height	30.5	34.7	28.9	32.6	67.4	63.7
Urban density	13.5	14.5	17.8	30.4	61.8	71.5
Road type	10.7	18.1	26.1	46.1	72.4	69.6
Road surface	75.6	80.2	58.7	58.4	95.0	95.0
Junction type	8.4	30.8	35.3	52.7	73.4	74.0
Green space	50.7	53.7	39.9	62.6	100.0	99.0
Amenity richness	13.5	12.6	17.8	25.4	53.4	53.0
Transit density	21.4	26.1	34.2	29.7	43.7	45.8
OVERALL (9)	27.8	33.4	32.9	41.3	54.9	54.9

Bold among externals = best external per row; **blue** = our trained Qwen3.5-9B sat (per-city / seen-unseen), same benchmark. Our trained model wins **every** task.

The gap: best zero-shot vs. our trained sat

task	best zero-shot	our trained sat	gap
Transit density	34.2	43.7	+9.5
Road surface	80.2	95.0	+14.8
Junction type	52.7	73.4	+20.7
Land use	53.4	75.5	+22.1
Road type	46.1	72.4	+26.3
Amenity richness	25.4	53.4	+28.0
Urban density	30.4	61.8	+31.4
Building height	34.7	67.4	+32.7
Green space	62.6	100.0	+37.4
OVERALL	41.3	54.9	+13.6

Smallest gap where a generic cue carries (road-surface texture, transit); largest where EOLLM-specific taxonomy grounding matters (green space, building height, urban density). Fine-tuning buys a large, consistent margin on the same satellite view.

- **EarthDial-4B-RGB is the strongest zero-shot RS-VLM** (41.3) — and the smallest (4B) and fastest. Its edge is **layout/structure** (road type, junction type, green space, urban density), where high-res InternVL tiling resolves geometry the 504px single-crop LLaVA models miss.
- **SkySenseGPT > GeoChat** throughout; both LLaVA models keep **road surface** (pure texture, view-agnostic) but collapse toward chance on count/context tasks.
- All externals are at/near 0.25 chance on **amenity richness** — counting services from one overhead tile is genuinely hard.

Caveat — don't over-read LHRs / EarthDial

LHRs-Bot-Nova and **EarthDial** share **OSM provenance** with our labels:

- LHRs-Bot pretrained on $\sim 1\text{M}$ **OSM-aligned** RS image–text pairs (thousands of cities).
- EarthDial's instruction QA is itself **OSM-derived**.

Our Family-1 labels are OSM-grounded, so their scores may reflect **shared OSM priors** (and possible image overlap), not pure visual skill — treat as an **upper bound**.

GeoChat/SkySenseGPT do *not* share this provenance.

Bottom line: even granting that upper bound, the best external (EarthDial 41.3) still trails our trained sat (54.9) by **13.6 pts** on the identical single satellite view — and our model wins all 9 tasks.

Run stats (reproducibility)

Model	n	unparseable	runtime	throughput	peak VRAM
GeoChat-7B	3135	0 (0.00%)	8.7 min	6.0 Q/s	40.2 GiB
SkySenseGPT-7B	3135	0 (0.00%)	8.7 min	6.0 Q/s	40.2 GiB
LHRS-Bot-Nova	3135	0 (0.00%)	2.4 min	21.5 Q/s	22.0 GiB
EarthDial-4B-RGB	3135	1 (0.03%)	1.9 min	27.5 Q/s	13.6 GiB

All via the shared `benchmark_suite` (byte-identical prompt + marked-sat tile). No topics skipped; single marked-sat tile per item. Unparseable counted as wrong. LHRS base LLM via the public `NousResearch/Meta-Llama-3-8B-Instruct` mirror (byte-identical to the gated Meta repo).